Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Ruff, Gary A. Ph.D.		
Project Title:	Secondary Fires: Initiation and Extinguishment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	garuff@nasa.gov	**Fax:**	FY 216-977-7065
PI Organization Type:	NASA CENTER	**Phone:**	216-433-5697
Organization Name:	Glenn Research Center		
PI Address 1:	MS 77-5		
PI Address 2:	21000 Brookpark Road		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135-3191	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	09/01/2001	**End Date:**	09/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	IH1011220		
Performance Goal No.:			
Performance Goal Text:	not received		
	The experiments are aimed directly at testing, understanding and improving NASA's existing policies and practices toward spacecraft fire safety. In this study, we examine several previously unaddressed issues regarding these fire safety practices and policies. Specifically, a. NASA Test 1 (an upward flame spread test) is the primary qualification test for materials' use on spacecraft. NASA Test 1, however, does not consider some possible fire sources and some of its assumptions remain unvalidated. These include, among others: a) Premixed fires can occur and heat and ignite solid materials, but these are unconsidered in Test 1; b) Solid materials may be heated well above normal spacecraft air temperature at the time of an accidental exposure to an ignition source (again unconsidered) and; c) The effect of firebrands in 1g is assumed to be worst case. b. Configuration control in microgravity is unvalidated. NASA requires that flammable materials be separated by 5 cm		

Task Description:	or more, so that fire from one material cannot ignite a neighboring material. c. There are also concerns with the fire suppression practices and policies. For example, there is uncertainty about the time to extinguish a fire upon termination of ventilation. The application of a jet of suppressant may itself produce firebrands from molten or charring material, and cause an accidental spread of fire. Finally, carbon dioxide, the suppressant of choice for the International Space Station, behaves differently than other diluent, in regard to its impact on the range of oxygen concentrations that will support a flame. The goal of the proposed research is to contribute to improved fire safety practices and policies for spacecraft and Martian habitat through the achievement of the following objectives: 1. Determine systematically the conditions that will ignite onboard flammable materials upon passage of an initial premixed gas, firebrand, or aerosol flame over these materials. 2. Test the effect of firebrands and configuration spacing. 3. Determine the effectiveness of the flow of CO2 extinguisher or other extinguishing agents. Fire is among the most feared hazards of human space flight. These experiments address fire safety practices and policies that have heretofore not been tested under conditions of spaceflight, i.e., in a microgravity environment. By performing parametric experiments both in normal gravity and reduced gravity on the KC-135 aircraft, as well as developing and analyzing related modeling, generality of the interpretation of the experimental findings will be pursued along with direct recommendations for fire safety practices and policies for fire safety on spacecraft and in Martian habitats. This is the principal value of the research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)